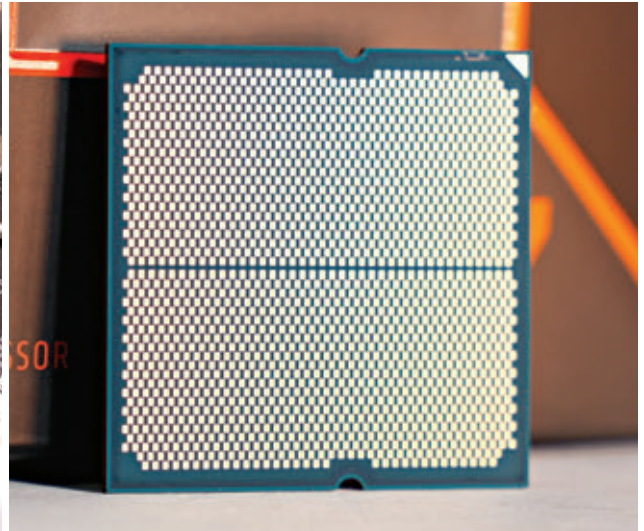
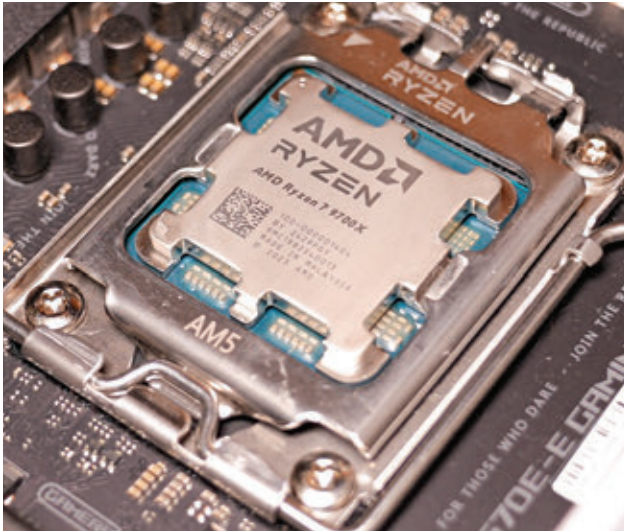




better in Microsoft Word and Excel. It even beats the 14600K in Word and Excel. Web Browsing benchmarks show pretty significant performance gains over the 7700X and the Intel 14600K. This is pretty good considering that Intel processors usually excel in these benchmarks.

AI benchmarks put the 9700X vastly ahead of the older gen 7950X3D and the 7700X. Given how a lot of applications have been including AI models as part of the workflow, this is a welcome change. Optimising for AI workloads is also something that server processors are doing a lot these days.

GAMING PERFORMANCE

If gaming dominates your workloads on the PC then the Ryzen 7 9700X is something you need not bother upgrading to from the 7700X. Titles like Cyberpunk 2077, Red Dead Redemption 2, and Forza Horizon 5 run at high frame rates, with no noticeable bottlenecking at 1440p and 4K resolutions. There's literally just 2-5 FPS difference between the two processors and on a few occasions, the older gen 7700X performed better than the 9700X. The Ryzen 7 7800X3D continues to dominate in this area and if we are to get a 9700X3D in the next couple of months, then gamers should hold out for that before deciding to upgrade, if at all you are one to upgrade between subsequent generations. Most folks tend to

wait a good 5-8 years before switching to a new platform.

POWER CONSUMPTION AND THERMALS

One area where the Ryzen 7 9700X stands out is in its power efficiency so we can say that there's certainly some decent headroom that can be leveraged with overclocking. In quite a few tests, the 9700X performed similar or slightly better compared to the 7700X at slightly lower power consumption levels. In testing, the CPU averaged around 87 Watts in multi-threaded benchmarks and much lower in benchmarks mimicking for most everyday tasks. Thermal management is effective too, provided you have a good cooler. The processor rarely exceeded 79°C under load, even during extended stress tests, which is a testament to the improvements made in the 4nm process. We switched out our AIO water cooler and put on an air cooler and the 9700X would still work totally fine without any throttling whatsoever.

VERDICT

The AMD Ryzen 7 9700X is a compelling option for gamers, content creators, and anyone looking for solid multi-core performance without breaking the bank. It slots neatly between the lower-end Ryzen 7 models and the more powerful Ryzen 9 series, offering excellent value for those who don't need the extreme

core counts of the latter but still want top-notch performance for gaming and productivity tasks. The reduced power consumption is certainly a great factor for someone looking to upgrade to a new platform. The competing 14th Gen processors consume a lot more power but they do perform a teensy bit better in some benchmarks. Considering the ongoing 'oxidation' issues, it would be wiser to opt for AMD processors for the time being. At INR 35,999, the Ryzen 7 9700X competes directly with Intel's Core i7 lineup, and while Intel may have a slight edge in single-core performance, AMD's offering here is more balanced overall, particularly in multi-threaded workloads and power efficiency. If you're building a new gaming or productivity rig and plan to pair your CPU with a powerful discrete GPU, the Ryzen 7 9700X should be high on your list of considerations. However, if you have an existing Ryzen 7 7000 series processor, then you're better off waiting for another generation or two.

—Mithun Mohandas

SPECIFICATIONS

Cores: 8 | Threads: 16 | Base clock: 3.8 GHz | Boost clock: 5.5 GHz | Process node: TSMC N4P | L3 Cache: 32 MB | L2 Cache: 1 MB per core | TDP: 65 W | Socket: AM5 | Architecture: Zen 5 "Granite Ridge"

CONTACT

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